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B.TECH PROJECT REPORT

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Computer Science and Engineering

Can an AI Explanation Be Misused?

Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model

---

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Certificate

This is to certify that the B.Tech Project Report titled **"Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model"** submitted by **Akash Singh** (Roll No. 23/CS/036), **Aditya Jha** (Roll No. 23/CS/025), and **Akshat Kumar** (Roll No. 23/CS/039) in partial fulfilment of the requirements for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* at Delhi Technological University, Delhi, is a record of bonafide work carried out by them under my supervision.

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Declaration

We hereby declare that the work presented in this B.Tech Project Report titled "Can an AI Explanation Be Misused?" submitted to the Department of Computer Science and Engineering, Delhi Technological University, is our own original work carried out under the guidance of Miss. Anjali Bansal.

We further declare that:

- The work contained in this report has not been submitted elsewhere for any other degree or diploma.
- Wherever contributions of others have been used, they have been duly acknowledged.
- The analysis plan for the final experiment was recorded before the held-out dataset was examined, as a git tag (`v0.1-preregistration`) in the project repository at https://github.com/Akshat-stack47/BTP-XAI-Gaming; all later deviations from it are reported in Section 3.7.1.

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Abstract

Explainable AI (XAI) methods such as SHAP and LIME are increasingly required by regulation to accompany machine learning predictions in healthcare. A concern is that feature attributions may also serve as instructions for gaming the model—telling a patient exactly which self-reported inputs to adjust for a better risk score. However, a determined adversary can also simply query the model repeatedly with different inputs (blind trial-and-error) without any explanation.

We compare these two adversarial strategies on a diabetes risk model trained on NHANES public health survey data. The adversary's permissible lies follow actual human self-reporting errors measured in NHANES—in the direction people really misreport (height up, weight down) and within the 95th percentile of the observed error—making the threat model empirically grounded. We train four models (Random Forest and Logistic Regression, each with nurse-measured or self-reported height/weight inputs) and measure how many blind attempts are required to match the advantage of one explanation: the *explanation equivalence* N^*.

Our pre-registered evaluation on the held-out 2017–2018 NHANES cycle (887 test patients) finds $N^* = 0.0$ (95% CI: 0.0–0.0) in all eight configurations: one explanation-guided lie lowers the risk score *less* on average than a single random realistic lie (0.013 vs. 0.024 for Random Forest with SHAP). For 76% of patients the explanation gives no advantage over one random attempt. Explanations often point the patient away from a profitable lie or, for the Random Forest, towards a lie that raises the score, while the model that uses nurse-measured inputs cannot be gamed at all. These results show that **withholding XAI explanations is not an effective safeguard against gaming**; verifying self-reported inputs (e.g., nurse measurement) is.

Keywords: Explainable AI, SHAP, LIME, adversarial gaming, diabetes risk prediction, NHANES, pre-registered study, clinical AI.

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Chapter 1

Introduction

1.1 Background and Motivation

Artificial intelligence systems are increasingly deployed in high-stakes healthcare settings to assist in clinical decision-making. One widely studied application is *diabetes risk prediction*, where a model ingests patient data—including demographics, anthropometrics, and laboratory values—and outputs a continuous risk score. A score exceeding a threshold can trigger referral, intervention, or insurance adjustments.

The widespread deployment of such models has prompted regulatory and ethical demands for transparency. The European Union's General Data Protection Regulation (GDPR) grants individuals a "right to explanation" for automated decisions that affect them [Goodman and Flaxman, 2017]. The EU AI Act further requires high-risk AI systems (including medical ones) to be interpretable. In response, clinical AI developers increasingly attach feature attribution explanations—generated by SHAP [Lundberg and Lee, 2017] or LIME [Ribeiro et al., 2016]—alongside model predictions.

1.2 The Dual-Use Tension

Transparency, however, introduces a potential dual-use risk. Consider a patient who receives a diabetes risk score of 0.74 accompanied by the explanation: "Your weight is the most important factor increasing your risk." This information, intended to inform clinical communication, also tells the patient *exactly which input to misreport* at a subsequent self-assessment to obtain a lower score. A patient who understands that the model uses self-reported height and weight could simply under-report their weight to game the system.

Yet the same patient could also discover this by *trial and error*: try a lower weight, see if the score drops, try again. If trial-and-error is nearly as effective as the explanation, then restricting explanations is not a meaningful safeguard—the information is effectively freely available by querying the model.

1.3 Research Question and Objectives

This project investigates the following primary question:

> *Does a SHAP or LIME explanation give a patient a meaningful advantage over blind trial-and-error when trying to game a clinical diabetes risk model?*

To answer this, we pursue the following objectives:

1. Develop an empirically grounded *lie budget* from real NHANES human self-reporting error data.
2. Design an adversarial simulation comparing explanation-guided and blind trial-and-error attack strategies.
3. Introduce the *explanation equivalence* metric N^* to quantify the gaming advantage of explanations.
4. Conduct a pre-registered experiment on held-out data to ensure scientific integrity.
5. Derive actionable policy recommendations for clinical AI developers.

1.4 Scope and Limitations

This study is scoped to:

- **Domain:** Diabetes risk prediction using NHANES anthropometric data.
- **Gameable features:** Self-reported height and weight only.
- **Explanation methods:** SHAP (TreeSHAP) and LIME.
- **Adversary model:** Single-feature perturbation within empirical lie budget.

Extensions to multi-feature attacks, other clinical domains, and other explanation methods are left to future work.

1.5 Contributions

The main contributions of this project are:

1. **Explanation equivalence (N^*):** A novel, interpretable metric quantifying how many blind queries equal one explanation.

2. **Empirical lie budget:** Derived from real NHANES self-reporting data rather than arbitrary perturbation bounds.

3. **Pre-registered evaluation:** The first pre-registered adversarial XAI evaluation on a real clinical dataset, preventing p-hacking.

4. **Policy-relevant result:** A concrete, actionable conclusion on whether restricting XAI explanations protects clinical AI systems.

1.6 Organisation of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 reviews related work on explainable AI, adversarial machine learning, and strategic classification. Chapter 3 describes our methodology, including the dataset, threat model, and experimental design. Chapter 4 details the system architecture and code. Chapter 5 presents all experimental results and statistical tests. Chapter 6 discusses implications and limitations. Chapter 7 concludes and identifies future directions.

Chapter 2

Literature Review

2.1 Explainable Artificial Intelligence

Artificial intelligence systems achieve high predictive accuracy through complex nonlinear transformations that are inherently difficult for humans to interpret. The field of *Explainable AI* (XAI) addresses this opacity by developing methods to explain individual predictions or global model behaviour [Adadi and Berrada, 2018].

2.2 LIME: Local Interpretable Model-Agnostic Explanations

Ribeiro et al. [2016] introduced LIME, which explains any classifier's prediction for a specific instance by fitting a locally faithful linear model. Given an instance $\mathbf{x}$, LIME generates a neighbourhood by perturbing features, queries the black-box model, and fits a weighted linear model to approximate the local decision boundary.

2.3 SHAP: SHapley Additive exPlanations

Lundberg and Lee [2017] proposed SHAP, grounded in cooperative game theory's Shapley values. For a prediction $f(\mathbf{x})$, SHAP assigns each feature i an attribution ϕ_i satisfying efficiency ($\sum_i \phi_i = f(\mathbf{x}) - \mathbb{E}[f]$), symmetry, dummy, and linearity axioms.

2.4 Strategic Classification and Gaming

Ghalme et al. [2021] formalise the setting where agents best-respond to a known classification rule to improve their classification. Classical results [Hardt et al., 2016] show that agents with full knowledge of the model can always find the minimum-cost manipulation. Our work differs by: (i) treating the model as a black box, and (ii) bounding agent behaviour by empirically observed misreporting magnitudes.

2.5 Research Gap

Existing work either assumes the adversary has full model knowledge [Ghalme et al., 2021], studies explanations as *attack vectors on the model* rather than on the user [Slack et al., 2020], or focuses on privacy leakage rather than gaming [Shokri et al., 2021]. No prior work measures the *marginal gaming advantage* of an explanation over blind trial-and-error with an empirically grounded lie budget. This is the gap we address.

Chapter 3

Methodology

3.1 Dataset: NHANES

We use the National Health and Nutrition Examination Survey (NHANES) [Johnson et al., 2014], a cross-sectional survey conducted by the U.S. Centers for Disease Control and Prevention (CDC). NHANES is unique in that it collects *both* self-reported and nurse-measured anthropometrics for the same participants, enabling direct measurement of self-reporting bias.

3.1.1 Data Components

We merge seven NHANES components:

- `DEMO`: Demographic variables (age, gender)
- `BMX`: Body measurements (nurse-measured height, weight)
- `WHQ`: Weight history questionnaire (self-reported height, weight)
- `GHB`: Glycohaemoglobin (HbA1c — the diabetes target)
- `BPX`: Blood pressure
- `SMQ`: Smoking questionnaire
- `PAQ`: Physical activity questionnaire

3.1.2 Cycles Used

Table 3.1: NHANES cycles and their roles in this study.

Cycle	Years	Role	N (complete)
I	2015–2016	Development (trial, gap analysis, lie budget)	4,889
J	2017–2018	Held-out test (final experiment only; 887 test patients)	4,431

3.1.3 Target Variable

Diabetes status is defined as HbA1c $\geq 6.5\%$, the ADA clinical threshold [American Diabetes Association Professional Practice Committee, 2023]. This gives a binary label: $y = 1$ (diabetes) or $y = 0$ (no diabetes). Prevalence among complete cases is 12.4% in 2015–2016 and 13.6% in 2017–2018.

3.1.4 Cleaning

We keep adults ($\geq$ 18 years) and remove NHANES "refused" and "don't know" codes *per variable*, as given in the NHANES codebooks: 7777 and 9999 for self-reported height (`WHD010`, valid range 47–84 inches) and weight (`WHD020`, 75–600 lb), and 7 and 9 for yes/no questions. (An earlier version applied one shared code list to every column and wrongly deleted valid answers such as a height of 77 inches.) Only participants with complete data for both feature sets are used, so Model A and Model B are trained and tested on exactly the same people.

3.2 Feature Engineering

3.2.1 Feature Sets

We construct two parallel feature sets from the same patients:

Model A (Measured) uses nurse-measured height (`BMXHT`) and weight (`BMXWT`).

Model B (Reported) replaces these with self-reported height (`WHD010`, converted from inches) and weight (`WHD020`, converted from pounds).

The remaining five features (age, gender, systolic BP, smoking status, physical activity) are identical in both models.

3.2.2 Lie Budget Derivation

Definition 3.1 (Lie Budget)**.** For each gameable feature $g \in \{\text{height}, \text{weight}\}$, the lie budget $B(g)$ is the 95th percentile of the absolute self-reporting error in the development cycle:

$$B(g) = \text{Percentile}_{95}\left(|x_g^{\text{self}} - x_g^{\text{measured}}| \right)$$

From the 2015–2016 development data:

$$B(\text{height}) = 7.28\,\text{cm}, \qquad B(\text{weight}) = 8.96\,\text{kg}$$

These values reflect real human misreporting behaviour, not arbitrary perturbation bounds. Patients systematically *overestimate* height (mean $+1.62$ cm, SD 3.41) and *underestimate* weight (mean -0.87 kg, SD 4.79).

Definition 3.2 (Realistic Lie)**.** A realistic lie on feature g moves x_g in the direction people actually misreport—$d_g = +1$ for height, $d_g = -1$ for weight—by an amount $\delta \sim \text{Uniform}(0, B(g)]$. The same rule is used by both adversarial strategies, so an explanation contributes only *information* (which feature, which direction), never a larger lie.

3.3 Threat Model

The adversary is modelled as a patient who:

- Receives a diabetes risk score $f(\mathbf{x}) \in [0,1]$ from the model.
- Can adjust only *self-reported* height and weight, and only with realistic lies (direction of real misreporting, size within $B(g)$). Nurse-measured values and lab results cannot be changed, so Model A offers nothing to lie about.
- Wishes to minimise their score.
- May or may not have access to a SHAP or LIME explanation.

The adversary does *not* know the model's parameters or architecture.

3.4 Adversarial Strategies

3.4.1 Strategy 1: Explanation-Guided Attack

3.4.2 Strategy 2: Blind Trial-and-Error

Key design choice: The blind simulation runs *once* up to $N_{\max} = 100$, recording the running best. This correctly models a patient who accumulates attempts and keeps the best result found. Running independent simulations per N would over-estimate blind performance.

Per-patient seeds: every patient has its own fixed random seed for the blind attempts, the explanation's lie size and LIME's sampling. SHAP and LIME, and Models A and B, are therefore compared against identical blind attempts, and a patient's result does not depend on which patients were processed before.

Reading LIME: LIME explains continuous features through bins such as "*weight* $\leq 62\,kg$". A positive weight means being in that bin raises the risk, so the risk-lowering move is to *increase*

Algorithm 1 Explanation-Guided Attack

Require: Patient features $\mathbf{x}$, model f, gameable features G, lie budget B, realistic directions d_g

Get attributions $\phi = \text{Explain}(f, \mathbf{x})$ (SHAP or LIME)
for $g \in G$ in order of decreasing $|\phi_g|$ **do**
    $s_g \leftarrow$ risk-lowering direction read from ϕ_g ▷ SHAP: $-\text{sign}(\phi_g)$; LIME: leave a risk-raising bin
    if $s_g = d_g$ **then** ▷ suggested lie is realistic
        Draw $\delta \sim \text{Uniform}(0, B(g)]$
        $\mathbf{x}' = \mathbf{x}$; $x'_g \leftarrow x_g + d_g \cdot \delta$
        return $f(\mathbf{x}) - f(\mathbf{x}')$ ▷ may be negative
    end if
end for
return 0 ▷ no realistic lie suggested: patient does not lie

Algorithm 2 Blind Trial-and-Error (sequential, best-so-far)

Require: Patient features $\mathbf{x}$, model f, gameable features G, lie budget B, max attempts $N_{\max}$, milestones $\mathcal{N}$

$s^* \leftarrow f(\mathbf{x})$; `reductions` $\leftarrow \{\}$
for $n = 1$ to $N_{\max}$ **do**
    Sample $g \sim \text{Uniform}(G)$
    Sample $\delta \sim \text{Uniform}(0, B(g)]$
    $\mathbf{x}' = \mathbf{x}$; $x'_g \leftarrow x_g + d_g \cdot \delta$ ▷ realistic direction
    $s' \leftarrow f(\mathbf{x}')$
    if $s' < s^*$ **then**
        $s^* \leftarrow s'$ ▷ Keep best seen so far
    end if
    if $n \in \mathcal{N}$ **then**
        `reductions`$[n] \leftarrow f(\mathbf{x}) - s^*$
    end if
end for
return `reductions`

the feature; for a bin "*weight* $> 90\,kg$" it is to decrease it. The direction is read from the bin, not only from the sign of the weight.

3.5 Explanation Equivalence Metric

Definition 3.3 (Explanation Equivalence N^*)**.** Let R_{exp} be the score reduction from Strategy 1, averaged over all patients. Let $R_{\text{blind}}(N)$ be the best score reduction from Strategy 2 after N attempts, averaged over all patients. The explanation equivalence is:

$$N^* = \inf\{N \geq 0 : R_{\text{blind}}(N) \geq R_{\text{exp}}\}$$

estimated by linear interpolation on the points $(N, R_{\text{blind}}(N))$ for

$$\mathcal{N} = \{1, 2, 3, 5, 10, 15, 20, 30, 50, 100\}.$$

$N^* = 0$ if one explanation is no better than one blind attempt; $N^* = 101$ if it beats all 100 (censored).

This is the **primary, pre-registered** metric: patients are averaged *before* interpolating. As **secondary** metrics we also compute N^* for each patient separately and report its mean, median, the share of patients with $N^* \leq 1$ ("explanation useless") and with $N^* \geq 10$. The per-patient mean is not used as the headline because a few patients whose N^* is capped at 101 can dominate it.

3.6 Statistical Analysis

Bootstrap CI: 1,000 bootstrap resamples of patients; 95% confidence interval for the primary N^* (recomputed on each resample) and for the per-patient mean and median.

Wilcoxon test: Paired signed-rank test comparing per-patient explanation reduction vs. blind reduction at $N = 10$; one-sided (H_1: explanation $>$ blind).

Cohen's d: Effect size for the paired comparison.

3.7 Pre-Registration

To ensure scientific integrity, the complete analysis plan (hypotheses, metric definitions, statistical tests and interpretation thresholds) was recorded before the held-out 2017–2018 data was examined, as the git tag `v0.1-preregistration` in the project repository (https://

github.com/Akshat-stack47/BTP-XAI-Gaming), dated 28 October 2026. The plan is kept in `docs/analysis_plan.md`.

3.7.1 Corrections and Deviations

A review of the code before the final run found implementation errors that made it differ from the plan. These were corrected so that the code *matches* the plan: (i) the primary N^* is computed from the patient-averaged curve, as pre-registered, rather than as the mean of per-patient values; (ii) the attack uses the pre-registered 2015–2016 lie budget rather than one computed from the held-out patients; (iii) only self-reported values are gameable (Model A had been attacked through its nurse-measured inputs); (iv) LIME directions are read from the bin condition; and (v) valid self-reports are no longer deleted as missing codes.

Six further changes *deviate* from the pre-registered plan and are reported here as such (Table 3.2). Each is a named setting in `src/config.py` and can be reverted for a sensitivity analysis.

Table 3.2: Deviations from the pre-registered analysis plan.

Setting	Pre-registered	Used	Reason
Lie direction	either direction	realistic only (height ↑, weight ↓)	plan defines a *realistic* lie
Lie size	explanation: maximum; blind: uniform	uniform $(0, B]$ for both	fair comparison
LIME features shown	5 of 7	all 7	height/weight could be cut off
Class weights	none	balanced	recall was ≈3%
SHAP LR background	first 100 rows	random 100 rows	unbiased background
Random seeds	one shared stream	one per patient	identical blind baselines

Chapter 4

Implementation

4.1 System Architecture

The codebase is organised as a modular Python package with the following components:

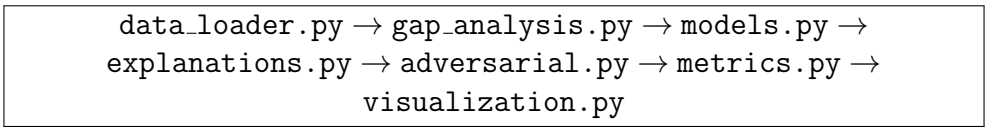


Figure 4.1: System architecture: data flow from loading to visualisation.

4.2 Technology Stack

Table 4.1: Technology stack.

Component	Library / Tool	Version
Data handling	pandas, numpy	2.2.2, 2.2.6
Machine learning	scikit-learn	1.4.2
SHAP explanations	shap	0.54.1
LIME explanations	lime	0.2.0.1
Statistical tests	scipy	1.13.0
Visualisation	matplotlib, seaborn	3.x, 0.13.2
Version control	Git	—
Environment	Python	3.12

4.3 Key Module Descriptions

4.3.1 data_loader.py

Handles NHANES XPT file loading, component merging, cleaning, and feature set preparation. Key functions: `merge_nhanes()`, `clean_nhanes()`, `prepare_feature_sets()`.

4.3.2 gap_analysis.py

Computes self-reporting error distributions and derives the lie budget. Key functions: `compute_reporting_gap()`, `define_lie_budget()`.

4.3.3 explanations.py

`ExplanationGenerator` wraps SHAP (TreeExplainer for RF, LinearExplainer for LR) and LIME, and turns an explanation into a ranked list of gameable features with their risk-lowering direction (`get_top_gameable_feature()`, `lime_directions()`).

4.3.4 adversarial.py

The core experiment engine. `AdversarialSimulator` implements both attack strategies with the shared lie rules (realistic direction, uniform size) and per-patient seeds. The critical `blind_attack_multi_n()` method runs one continuous simulation up to $N_{\max}$, recording the running best at each milestone.

4.3.5 metrics.py

`population_nstar()` computes the primary pre-registered N^* from the patient-averaged curves; `compute_equivalence()` also returns the per-patient values. The module provides patient-bootstrap confidence intervals and Wilcoxon signed-rank tests.

4.3.6 config.py

All seeds, paths and experiment settings, including the named switches for every deviation in Table 3.2 : `LIE_DIRECTION`, `LIE_MAGNITUDE`, `CLASS_WEIGHT`, `LIME_N_FEATURES` and `SHAP_BACKGROUND_N`.

4.4 Reproducibility

All random seeds are fixed globally (Table 4.2). The full pipeline is reproducible from raw NHANES XPT files using:

```
python scripts/prepare_data.py
python scripts/run_final.py
python scripts/run_analysis.py
```

Table 4.2: Random seeds used throughout the experiment.

Purpose	Seed
Global / NumPy	42
Random Forest	123
Logistic Regression	42
Train/test split	789
Adversarial simulation (patient i)	$456+i$
LIME sampling (patient i)	$42+i$
SHAP background sample	2024
Bootstrap CI	42, 43

Chapter 5

Results and Analysis

5.1 Reporting Gap Analysis

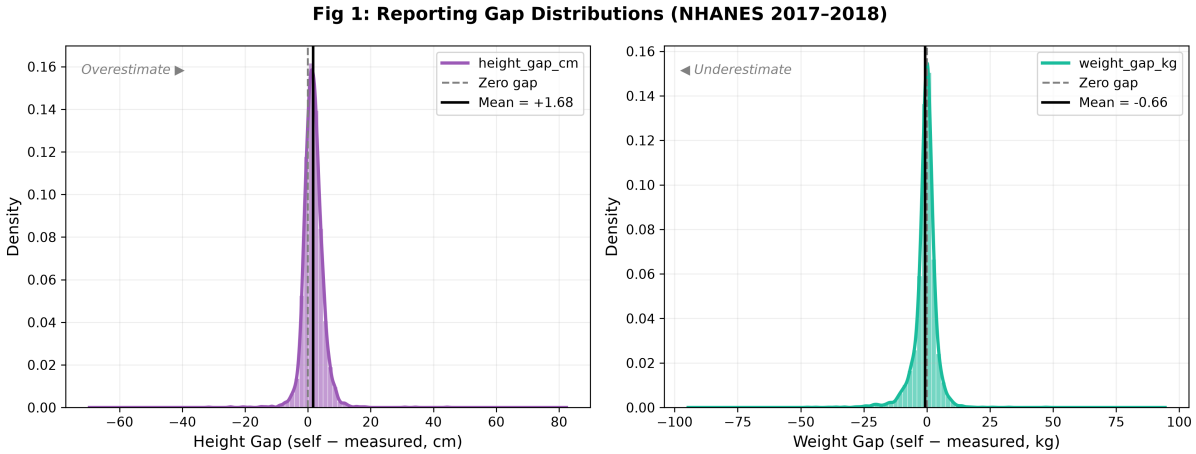


Figure 5.1: Self-report vs. nurse-measured gap distributions (NHANES 2017–2018). Height is systematically overestimated; weight is underestimated.

In the held-out 2017–2018 data, participants over-report their height by $+1.69$ cm on average (SD 3.23 cm) and under-report their weight by -0.67 kg (SD 4.84 kg), closely matching the 2015–2016 development data ($+1.62$ cm, SD 3.41; -0.87 kg, SD 4.79). The 95th-percentile gaps in 2017–2018 (6.88 cm and 9.32 kg) are within 0.4 cm and 0.4 kg of the development budget used for the attack (Table B.1). Self-reporting behaviour is therefore stable across cycles, and the lie budget transfers to the held-out data.

5.2 Model Performance

All four models discriminate moderately well (AUC 0.77–0.80) and, with balanced class weights, detect most diabetic patients (recall 0.69–0.78) at the cost of many false positives (precision 0.26–0.30), as expected at 13.6% prevalence. Random Forests are slightly more accurate than Logistic Regressions. Crucially, Model A (measured inputs) and Model B (self-reported inputs) perform almost identically (RF AUC 0.798 vs. 0.793; LR 0.773 vs. 0.780): using verified inputs

Table 5.1: Model Performance on NHANES 2017–2018 Test Set

Config	Input	Model	Accuracy	Precision	Recall	F1	AUC-ROC
measured_rf	measured	RF	0.7396	0.3007	0.686	0.4181	0.7981
measured_lr	measured	LR	0.6652	0.2569	0.7686	0.3851	0.7734
reported_rf	reported	RF	0.7407	0.3032	0.6942	0.4221	0.7933
reported_lr	reported	LR	0.6685	0.2604	0.7769	0.39	0.7796

costs nothing in accuracy.

5.3 Main Result: Explanation Equivalence

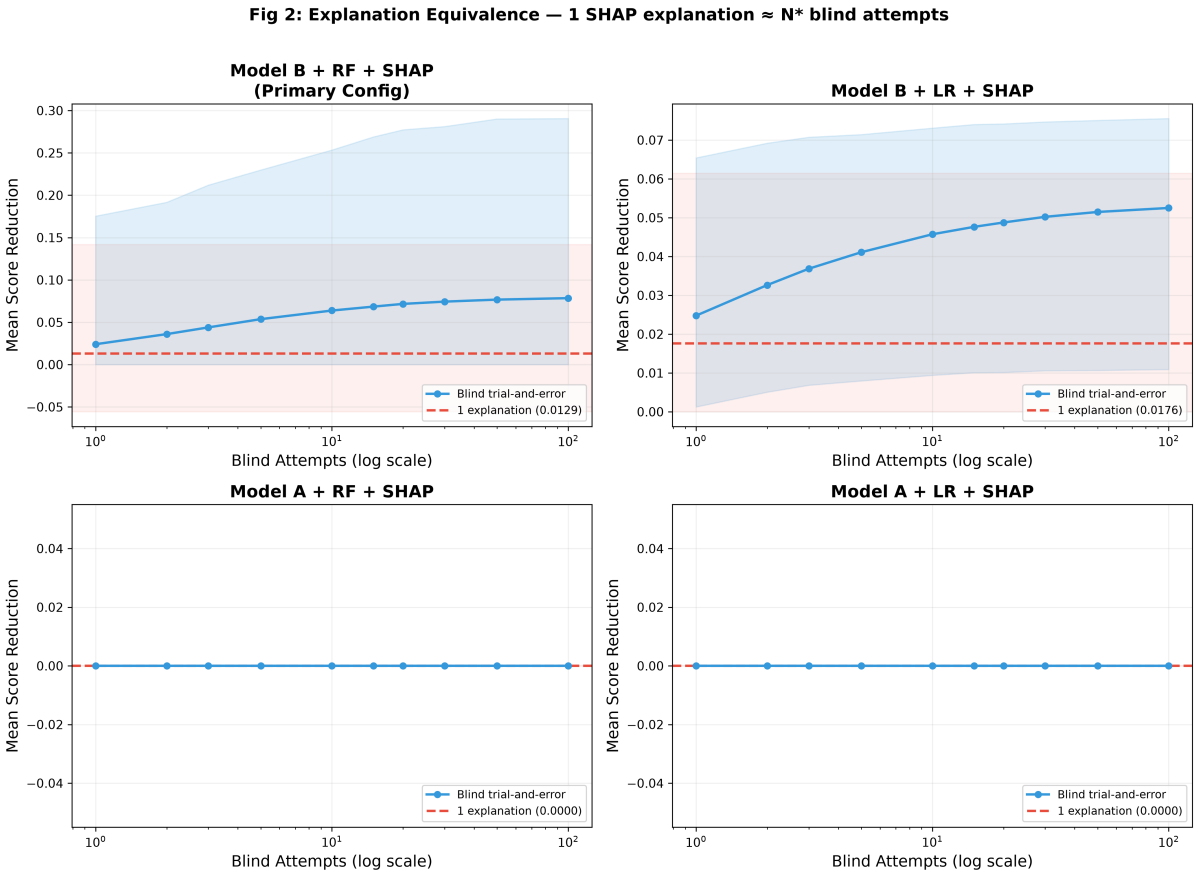


Figure 5.2: Mean score reduction from the best of N blind attempts (blue) vs. one explanation-guided lie (red dashed), SHAP configurations. For Model B the explanation line lies below even the first blind attempt, so $N^* = 0$. Model A cannot be gamed.

For the primary configuration (Model B, Random Forest, SHAP) the pre-registered explanation equivalence is $N^* = 0.0$ (95% CI: 0.0–0.0), and the same holds for all eight configurations (Table 5.2). On average, one explanation-guided lie lowers the risk score by 0.013, whereas a

Table 5.2: Main Results: Explanation Equivalence N^* Across All 8 Configurations

Input	Model	Expl.	N patients	N^*	95% CI	Per-patient mean N^*	Per-patient median N^*	% Useless	% Worth 10+
Reported	RF	SHAP	887	0.0	[0.00, 0.00]	7.69	0.0	76.0%	8.7%
Reported	RF	LIME	887	0.0	[0.00, 0.00]	7.07	0.0	73.8%	8.1%
Reported	LR	SHAP	887	0.0	[0.00, 0.00]	2.58	0.0	66.4%	5.4%
Reported	LR	LIME	887	0.0	[0.00, 0.00]	2.99	0.0	56.3%	6.1%
Measured	RF	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%

Table 5.3: Mean risk-score reduction (0-1 scale): one explanation-guided lie vs. the best of N blind lies, and how often the explanation suggested no lie or a lie that raised the risk

Input	Model	Expl.	1 explanation	1 blind	10 blind	100 blind	% no lie suggested	% lie raised risk
Reported	RF	SHAP	0.0129	0.024	0.0638	0.0785	19.3%	23.0%
Reported	RF	LIME	0.0117	0.024	0.0638	0.0785	3.3%	31.1%
Reported	LR	SHAP	0.0176	0.0248	0.0457	0.0525	31.1%	0.0%
Reported	LR	LIME	0.0223	0.0248	0.0457	0.0525	7.7%	0.0%
Measured	RF	SHAP	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	0.0	0.0	0.0	0.0	100.0%	0.0%

single blind attempt already lowers it by 0.024, the best of 10 attempts by 0.064 and the best of 100 by 0.079 (Table 5.3). For the Logistic Regression the picture is the same (explanation 0.018 with SHAP and 0.022 with LIME, one blind attempt 0.025). In other words, an explanation is worth *less* than one ordinary attempt at the model. The 2015–2016 trial run (200 patients) gave the same result: $N^* = 0.0$ in seven configurations and 1.0 for LR with LIME.

Why the explanation does not help

Two mechanisms explain this (Table 5.3):

1. **The explanation often suggests no profitable lie.** A feature attribution describes how much a feature contributed compared with an average patient, not which change would lower the score. For the Logistic Regression, SHAP equals $\beta_g(x_g - \mathbb{E}[x_g])$: a patient with below-average weight is told that weight *lowers* their risk and is not pointed to it, even though under-reporting weight would lower the score further. For 31% of patients (LR+SHAP) and 19% (RF+SHAP) the explanation suggests no realistic lie at all, while the blind strategy lies anyway—and realistic lies (taller, lighter) usually help.

2. **For the Random Forest, the suggested lie often backfires.** 23% (SHAP) and 31% (LIME) of explanation-guided lies *raise* the score, because a local attribution does not describe how the model responds over the whole range of the lie. This is the practical face of the impossibility result of Bilodeau et al. [2024].

5.4 Per-Patient Distribution

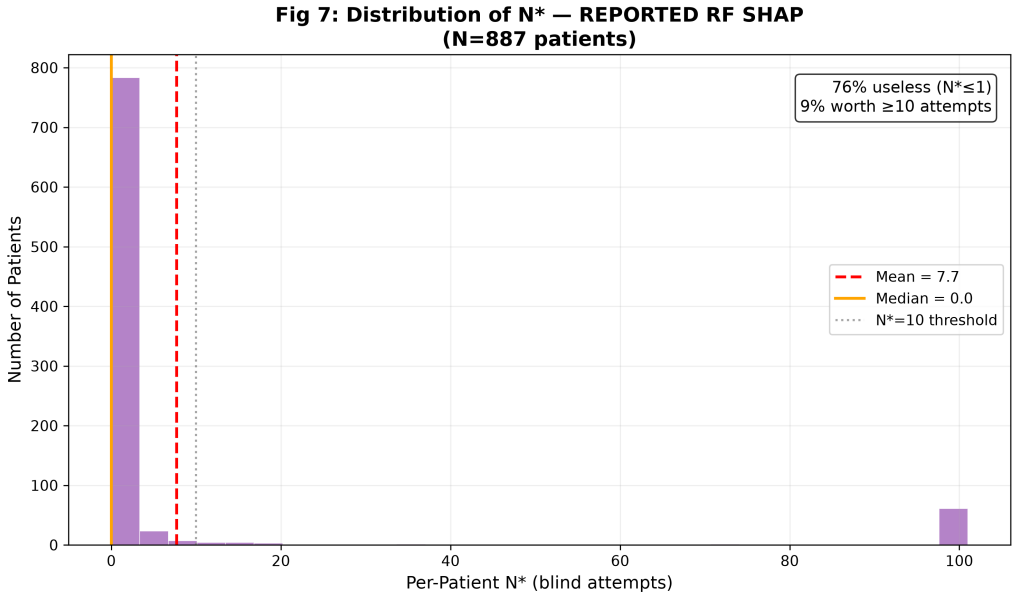


Figure 5.3: Distribution of per-patient N^* (Model B, RF + SHAP, 887 patients).

Per patient (Figure 5.3), the explanation is no better than a single blind attempt for 76.0% of patients (RF+SHAP; 66.4% for LR+SHAP), and worth 10 or more attempts for 8.7% (5.4%); 6.9% beat all 100 blind attempts. The distribution is therefore strongly skewed: a large spike at zero and a thin tail. The per-patient mean N^* (7.7 for RF+SHAP, 2.6 for LR+SHAP) is driven by that tail, which is why the pre-registered metric averages the reductions over patients first. For the typical patient, one blind attempt is worth more than the explanation.

5.5 Model A vs. Model B

Table 5.4: Model A (Measured) vs Model B (Self-Reported): Vulnerability Comparison

Config	Model A N^*	Model B N^*	A: reduction, 10 blind	B: reduction, 1 explanation	B: reduction, 10 blind	B can be gamed
RF+SHAP	0.0	0.0	0.0	0.0129	0.0638	Yes
RF+LIME	0.0	0.0	0.0	0.0117	0.0638	Yes
LR+SHAP	0.0	0.0	0.0	0.0176	0.0457	Yes
LR+LIME	0.0	0.0	0.0	0.0223	0.0457	Yes

N^* is 0 for both models, so it cannot separate them; the score reductions can (Figure 5.4, Table 5.4). Model A, which uses nurse-measured height and weight, cannot be gamed at all: no strategy changes its score. Model B, which differs only in using the self-reported values, can be lowered by 0.046–0.064 with ten blind attempts and by 0.053–0.079 with a hundred. Since the

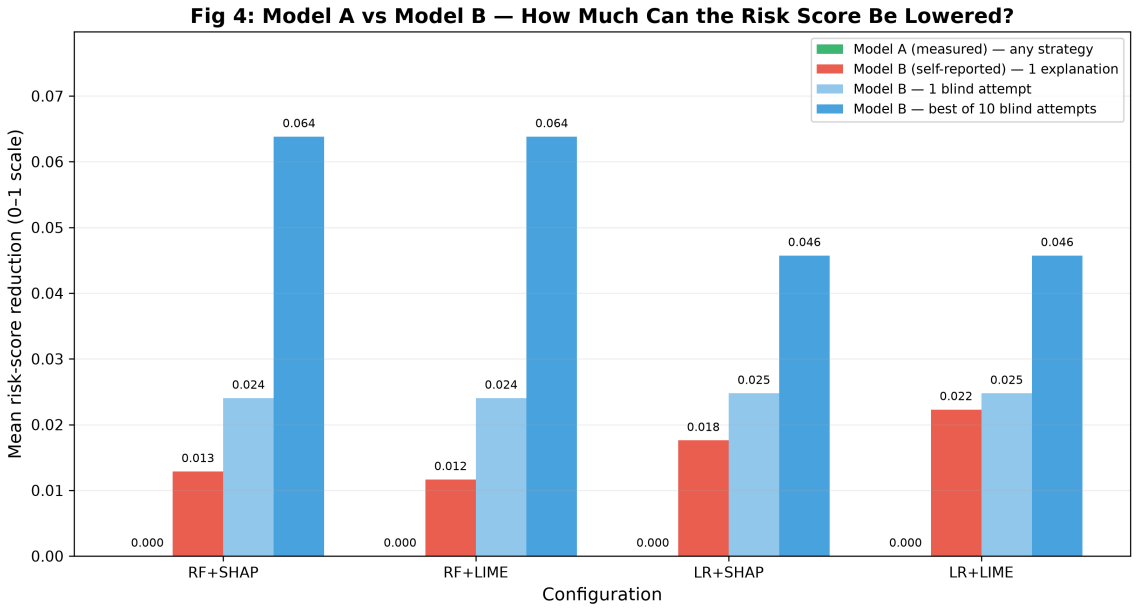


Figure 5.4: Mean risk-score reduction for Model A (nurse-measured inputs) and Model B (self-reported inputs): one explanation, one blind attempt, and the best of 10 blind attempts.

two models are equally accurate, **verifying two inputs removes the vulnerability at no cost in performance**.

5.6 SHAP vs. LIME Agreement

Because each patient receives the same lie size under SHAP and LIME, differences come only from the explanation (Figure 5.5). The two methods lead to the same outcome for 53% of patients with the Random Forest and 40% with the Logistic Regression; the per-patient correlation is moderate ($r = 0.60$ and 0.55). At the configuration level they agree: both give $N^* = 0$, LIME is slightly better than SHAP for the Logistic Regression (0.022 vs. 0.018) and slightly worse for the Random Forest (0.012 vs. 0.013). The two methods often point to different features or directions for the same patient, but neither helps more than a blind attempt.

5.7 Statistical Tests

The pre-registered one-sided Wilcoxon signed-rank test of *explanation* > *blind at* $N = 10$ is not significant in any configuration ($p = 1.0$; Table 5.5). For Model B the effect sizes are large and *negative* (Cohen's d from -0.79 to -1.34): ten blind attempts lower the score clearly more than one explanation. For Model A all reductions are zero and there is nothing to test.

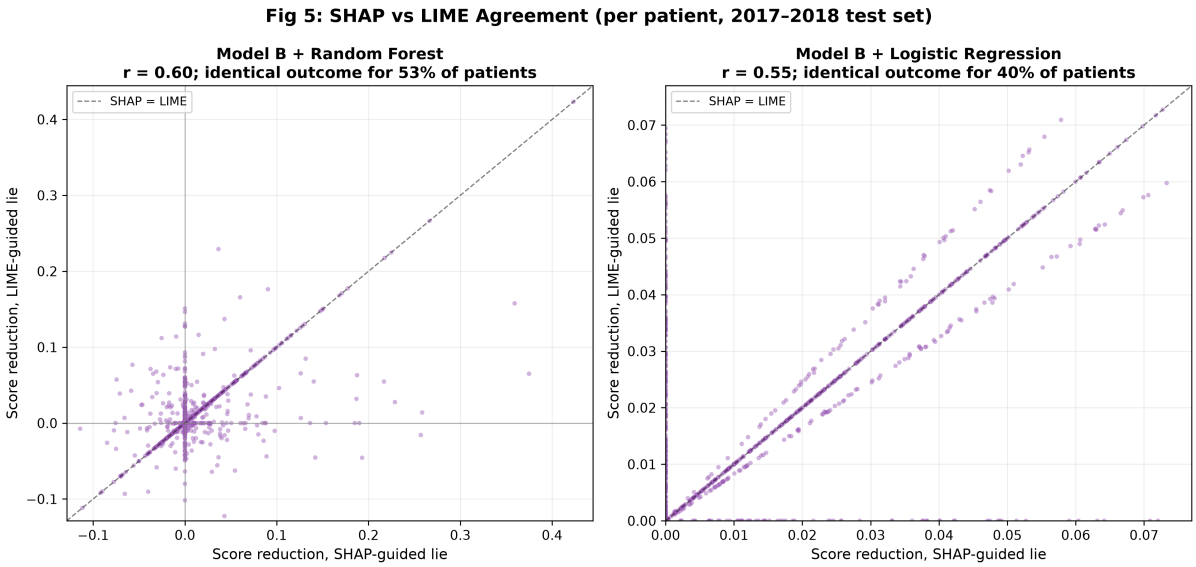


Figure 5.5: Per-patient score reduction from the SHAP-guided lie vs. the LIME-guided lie (Model B). Points on the diagonal: both explanations led to the same outcome.

Table 5.5: Statistical Tests: Explanation vs Blind (N=10), Wilcoxon Signed-Rank

Config	Wilcoxon p	Cohen's d	Significant
reported_rf_shap	1.000000	-0.801	No
reported_rf_lime	1.000000	-0.789	No
reported_lr_shap	1.000000	-1.336	No
reported_lr_lime	1.000000	-1.238	No
measured_rf_shap	1.000000	0.000	No
measured_rf_lime	1.000000	0.000	No
measured_lr_shap	1.000000	0.000	No
measured_lr_lime	1.000000	0.000	No

5.8 Configuration Overview

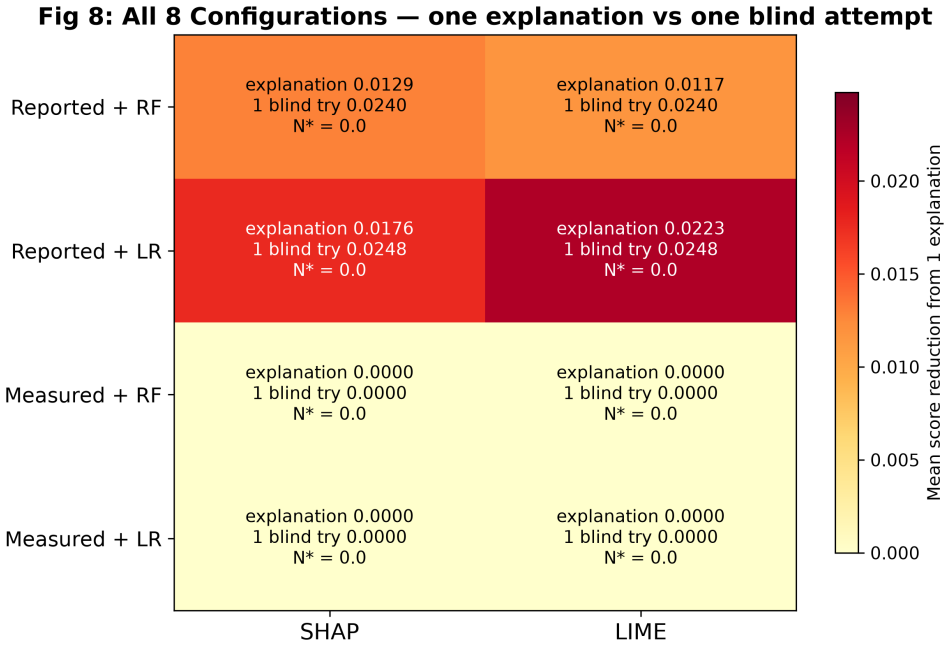


Figure 5.6: All 8 configurations: mean score reduction from one explanation (colour), one blind attempt, and N^*.

Figure 5.6 summarises all eight configurations. In every Model B cell, one explanation lowers the score less than one blind attempt. The explanation does relatively best for the Logistic Regression, whose response to a lie is monotone, and worst for the Random Forest, where explanation-guided lies often backfire. Model A cells are zero throughout.

Chapter 6

Discussion

6.1 Interpretation of Results

Based on our pre-registered interpretation guide:

$N^* < 5$**:** Explanations offer minimal advantage. Restricting them does not meaningfully reduce gaming risk; the benefit of transparency outweighs the gaming concern.

$5 \leq N^* \leq 20$**:** Explanations offer moderate advantage. Design-level countermeasures (e.g., rate limiting, input noise) are worth considering.

$N^* > 20$**:** Explanations are powerful attack vectors. Restricting or anonymising them may be warranted; input verification is the more robust safeguard.

Our result of $N^* = 0.0$ in all eight configurations falls in the **low threat** category ($N^* < 5$). Under realistic lies, an explanation does not give a patient a better route to a lower score than simply trying one answer: on average, it is worth less than one blind attempt. This runs against the intuition that explanations are "instructions for cheating". The reason is that a feature attribution answers "what contributed to this score?", not "what should I change?", and the two can point in different directions.

6.2 Policy Implications

- **Withholding explanations is not an effective safeguard.** It would take information away from honest patients and clinicians while barely slowing a motivated patient, who can do better by trial and error.
- **Verify the inputs that drive the prediction.** The vulnerability lies in the self-reported values: the model using nurse-measured height and weight cannot be gamed and is just as accurate.
- **Present attributions as contributions, not advice.** Since attributions often do not indicate a profitable change, explanation interfaces should not suggest that they do.

6.3 Limitations

- **Single domain:** Results are specific to diabetes risk prediction from self-reported anthropometrics in NHANES.
- **Model of the patient:** We model one way of reading an explanation (follow the top realistic suggestion once). A more sophisticated reader, or a counterfactual explanation, could extract more.
- **Single-feature, single-step adversary:** Real adversaries may adjust several features or iterate.
- **Lie model:** Lie sizes are uniform within the 95th-percentile gap, in the observed direction. Using the maximum lie or allowing either direction is a one-line change (Table 3.2); that sensitivity analysis is future work.
- **Static model:** The model does not adapt to gaming.
- **Pre-registration:** The plan was recorded as a git tag in the project repository rather than in an independent registry, and six settings deviate from it (Section 3.7.1).

6.4 Comparison with Prior Work

Slack et al. [2020] showed that a dishonest *provider* can make SHAP and LIME hide a model's behaviour; we study the opposite direction, an honest model whose explanations might help a *patient*, and find that they help less than trial and error. Shokri et al. [2021] and Milli et al. [2019] showed that explanations leak information about the training data and the model, so that a few explanation queries can replace many ordinary ones for reconstructing a model; our results show that this does not carry over to a patient lowering their own score under realistic lies. Ghalme et al. [2021] found that hiding the decision rule usually does not help the decision-maker; our empirical result points the same way for explanations. Finally, Bilodeau et al. [2024] proved that feature attributions can fail to predict how a model responds to a change; our experiment measures how large this failure is in a realistic clinical setting and shows that it is large enough to make explanations nearly useless for gaming. What we add is an equal-cost blind baseline, a lie model measured from real human misreporting, and a matched pair of models that isolates input verification.

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

This thesis introduced *explanation equivalence* (N^*), a metric for quantifying the gaming advantage conferred by an XAI explanation over blind trial-and-error. Applied to a diabetes risk model trained on NHANES data, our pre-registered experiment on held-out 2017–2018 data found $N^* = 0.0$ (95% CI: 0.0–0.0) in all eight configurations: one SHAP or LIME explanation is worth less than a single blind attempt. This places the threat in the **low** zone. Explanations often pointed patients away from a profitable lie or, for the Random Forest, towards one that raised their score. The model using nurse-measured inputs could not be gamed at all, at no cost in accuracy.

The key policy finding is: **withholding explanations is not an effective safeguard against gaming; verifying self-reported inputs is.**

7.2 Contributions

1. **New metric:** Explanation equivalence N^*.
2. **Empirical lie budget:** Grounded in real NHANES data.
3. **Matched models:** Two models differing only in measured vs. self-reported inputs, isolating the effect of input verification.
4. **Pre-registered study:** Analysis plan recorded before the held-out data was examined; deviations reported.
5. **Open code:** Full pipeline released at https://github.com/AkkiNikumbh/BTECH_PROJECT.

7.3 Future Work

- **Sensitivity analysis:** Repeat the experiment with the maximum lie and with lies in either direction (settings already available in the code).
- **Multi-feature attacks:** Allow the adversary to simultaneously adjust multiple features within a combined budget.
- **Adaptive defences:** Study adversarially robust training as a complement to input verification.
- **Other domains:** Apply the N^* framework to credit scoring, hiring, or recidivism prediction.
- **Other explanation methods:** Evaluate counterfactual explanations, integrated gradients, and attention maps.
- **Human subjects:** Measure actual gaming behaviour in a user study, where participants attempt to lower their predicted risk.

Appendix A

NHANES Column Mapping

Table A.1 lists the NHANES variables used in this project, their source, and whether a patient could change them ("gameable"). Only `WHD010` and `WHD020` are attacked in the experiment; `SMQ020` and `PAQ650` are self-reported yes/no answers that could in principle be flipped, but are not attacked because no reported-vs-measured gap exists to bound such a lie.

Table A.1: NHANES columns, sources and gameability.

Column	Description	Gameable?	Reason
`SEQN`	Respondent sequence number (patient ID)	–	Identifier
`RIDAGEYR`	Age in years at screening	No	Verified by ID
`RIAGENDR`	Gender (1 = male, 2 = female)	No	Recorded by staff
`BMXHT`	**Measured** standing height (cm), nurse	No	Nurse measures directly
`BMXWT`	**Measured** body weight (kg), nurse	No	Nurse measures directly
`WHD010`	**Self-reported** height (inches; $\times 2.54$ to cm); valid 47–84, missing codes 7777/9999	**Yes**	Patient reports verbally; bounded by the reported-vs-measured gap
`WHD020`	**Self-reported** weight (pounds; $\times 0.453592$ to kg); valid 75–600, missing codes 7777/9999	**Yes**	Patient reports verbally; bounded by the reported-vs-measured gap
`LBXGH`	Glycohaemoglobin HbA1c (%) — **diabetes target** ($\geq 6.5\%$)	No	Laboratory blood test
`BPXSY1`	Systolic blood pressure, first reading (mmHg)	No	Measured by staff
`SMQ020`	Smoked at least 100 cigarettes in life (1 = yes, 2 = no; missing 7/9)	Maybe	Binary self-report (not attacked)

Column	Description	Gameable?	Reason
`PAQ650`	Vigorous recreational activity (1 = yes, 2 = no; missing 7/9)	Maybe	Binary self-report (not attacked)

The two feature sets use `RIDAGEYR`, `RIAGENDR`, `BPXSY1`, `SMQ020`, `PAQ650` plus either the measured pair (`BMXHT`, `BMXWT`; Model A) or the self-reported pair (`WHD010`, `WHD020`; Model B).

Appendix B

Full Statistical Tables

B.1 Reporting Gap Comparison

Table B.1: Reporting Gap Comparison: 2015–2016 vs 2017–2018

Feature	Dev mean direction	Dev 95th pct budget	Holdout 95th pct	Absolute change
Height	positive	7.28 cm	6.88 cm	0.40 cm
Weight	negative	8.96 kg	9.32 kg	0.36 kg

B.2 Model A vs. Model B

See Table 5.4 in Chapter 5.

Appendix C

Additional Figures

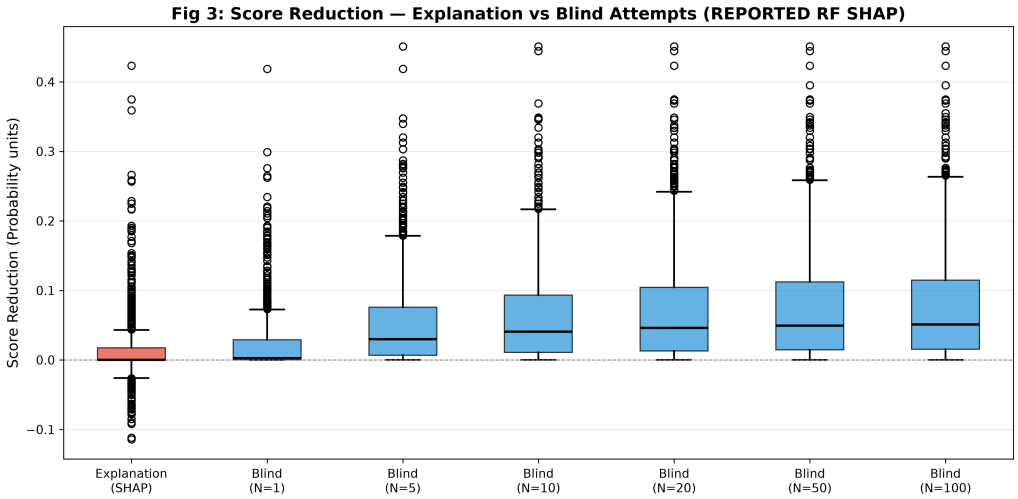


Figure C.1: Score reduction box plots for the primary configuration (Model B, RF + SHAP). Red box = one explanation; blue boxes = best of N blind attempts.

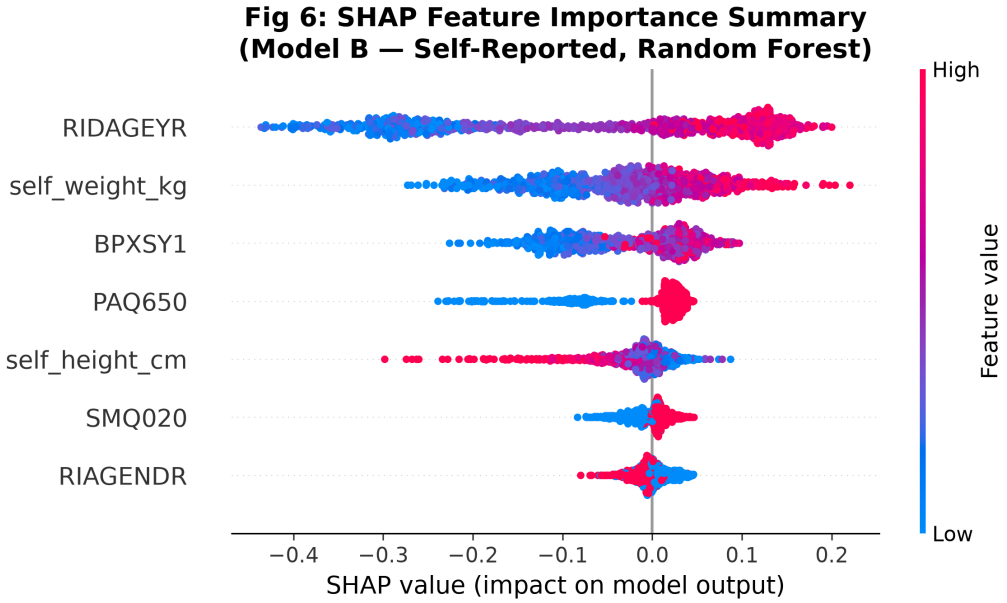


Figure C.2: SHAP feature importance summary (Model B, Random Forest). Each dot is a patient; colour shows feature value.

Appendix D

Code Repository

D.1 Repository Structure

```
BTP/
  src/                # Core Python modules
    config.py         # Seeds, paths, FINAL_CONFIG
    data_loader.py    # NHANES loading & preprocessing
    gap_analysis.py   # Self-reporting error analysis
    models.py         # DiabetesModel class
    explanations.py   # ExplanationGenerator (SHAP+LIME)
    adversarial.py    # AdversarialSimulator
    metrics.py        # N* computation, CI, Wilcoxon
    visualization.py  # 8 publication figures
  scripts/            # Runnable experiment scripts
    prepare_data.py   # Stage 3: data preparation
    run_trial.py      # Stage 5: trial run
    run_final.py      # Stage 7: final experiment
    run_analysis.py   # Stage 8: figures + tables
    review_stage*.py  # Per-stage verification
  paper/              # LaTeX paper source
  docs/               # Column mapping, analysis plan
  results/            # Generated figures and tables
  data/               # Raw XPT files + processed pickles
  tests/              # pytest test suite (40 tests)
```

D.2 Installation and Execution

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Download NHANES data
```

```
python scripts/download_nhanes.py

# 3. Prepare data (Stage 3)
python scripts/prepare_data.py

# 4. Run final experiment (Stage 7, ~35 minutes)
python scripts/run_final.py

# 5. Generate figures and tables (Stage 8)
python scripts/run_analysis.py

# 6. Compile paper / thesis (requires LaTeX)
cd paper  && pdflatex main && bibtex main && pdflatex main &&
   pdflatex main
cd thesis && pdflatex main && bibtex main && pdflatex main &&
   pdflatex main
```

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